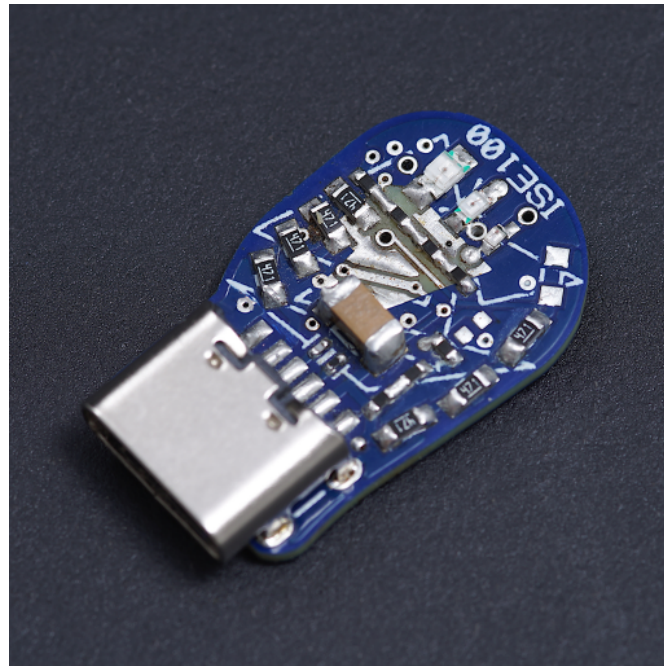

ISE-100



INSTRUCTIONS

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1 Risk Assessment and Disposal

1.1 Risk Assessment

This kit is not suitable for children under 10 years of age. Assembly and use should always be supervised by an adult. Please note the following risks:

- Small parts can be swallowed. Keep the kit out of reach of small children.
- Tools such as tweezers or side cutters can cause injuries.
- The soldering iron becomes very hot and can cause burns.
- Use a fireproof surface to protect work areas.
- Flux can splatter during soldering. Wear protective goggles.
- Solder fumes can be harmful to health. Ensure proper ventilation.
- Small components can be easily lost. Work on a bright, tidy surface.
- Mishandling can result in damage to components.

1.2 Disposal Instructions

We care about the environment! Here are some tips on how to dispose of our kit and its packaging responsibly:

- Electronic components and batteries must not be disposed of in household waste. Please take them to a collection point or recycling center.
- Our packaging is made of environmentally friendly materials. Recycle it in the appropriate bins.
- Separate plastic, paper, and other materials to facilitate recycling.



2 Basics

2.1 Soldering

Our soldering exercises are designed to teach you the basics of soldering step by step. Don't worry, no one becomes an expert overnight! Just like in many things, *"practice makes perfect."*

You can find a helpful video about common soldering mistakes on our YouTube channel:

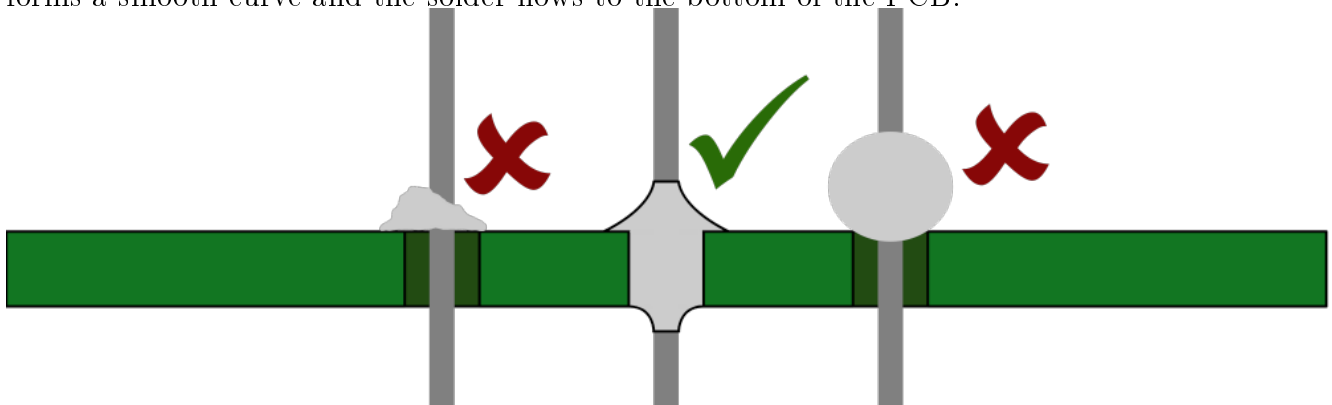
<https://youtu.be/CPXZM8r8xFw>

Here are some important points in brief:

- The ideal **temperature** for the soldering iron should be around **350°C**. It is recommended to use lead-free solder.
- The choice of soldering tip should match the size of the component. A **1.6 mm chisel tip** is ideal for most solder joints – it works for SMD components of size 0805 and wires up to 1.5 mm².
- To protect yourself from solder splashes or flying wires when cutting, it's recommended to wear **safety glasses**.
- Use a **heat-resistant surface** to avoid burn marks on your table.

The right amount of solder is crucial. Make sure the solder covers both the component and the PCB. The soldering iron must touch both the **component** and the **PCB** at the same time so that both are heated properly.

The following image shows typical solder joints: On the left, too little solder was used, while on the right, a ball forms – this happens when only the wire is heated. A perfect solder joint (middle) forms a smooth curve and the solder flows to the bottom of the PCB.



3 ISE-100

3.1 Bill of Materials

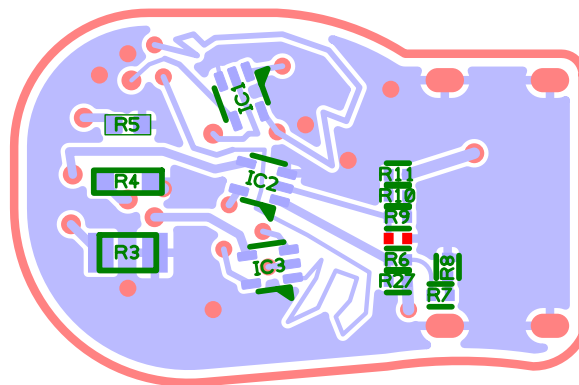
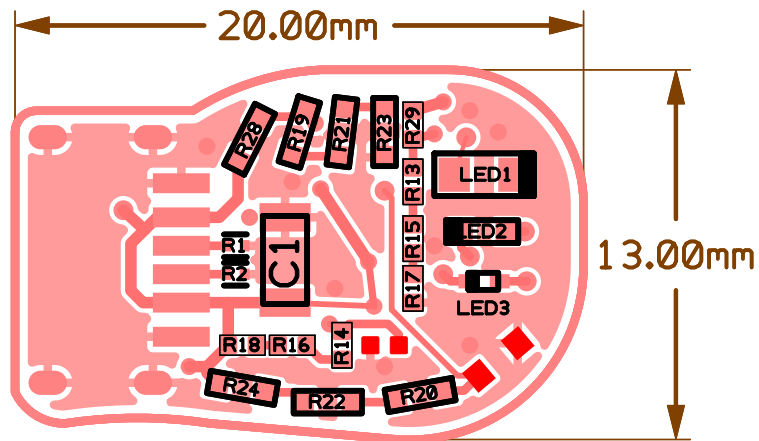
The following components are required for assembly. Refer to the schematic and assembly plan to determine where each component should be placed:

Name	Value	Qty.	Part
C1	4p7	1	C_Kerko_1206_4p7
IC1	AND	3	SN74LVC1G08DRLR
IC2			
IC3			
LED1		1	LED_0805_generic
LED2		1	LED_0603_generic
LED3		1	LED_0402_generic
R1	5k1	10	R_0201_generic
R2			
R6			
R7			
R8			
R9			
R10			
R11			
R12			
R27			
R3	470R	1	R_0805_generic
R4	470R	9	R_0603_generic
R19			
R20			
R21			
R22			
R23			
R24			
R25			
R28			
R5	470R	9	R_0402_generic
R13			
R14			
R15			
R16			
R17			
R18			
R26			
R29			
X1	USB-C-PD	1	632722100213

Table 1: Bill of Materials for Assembly

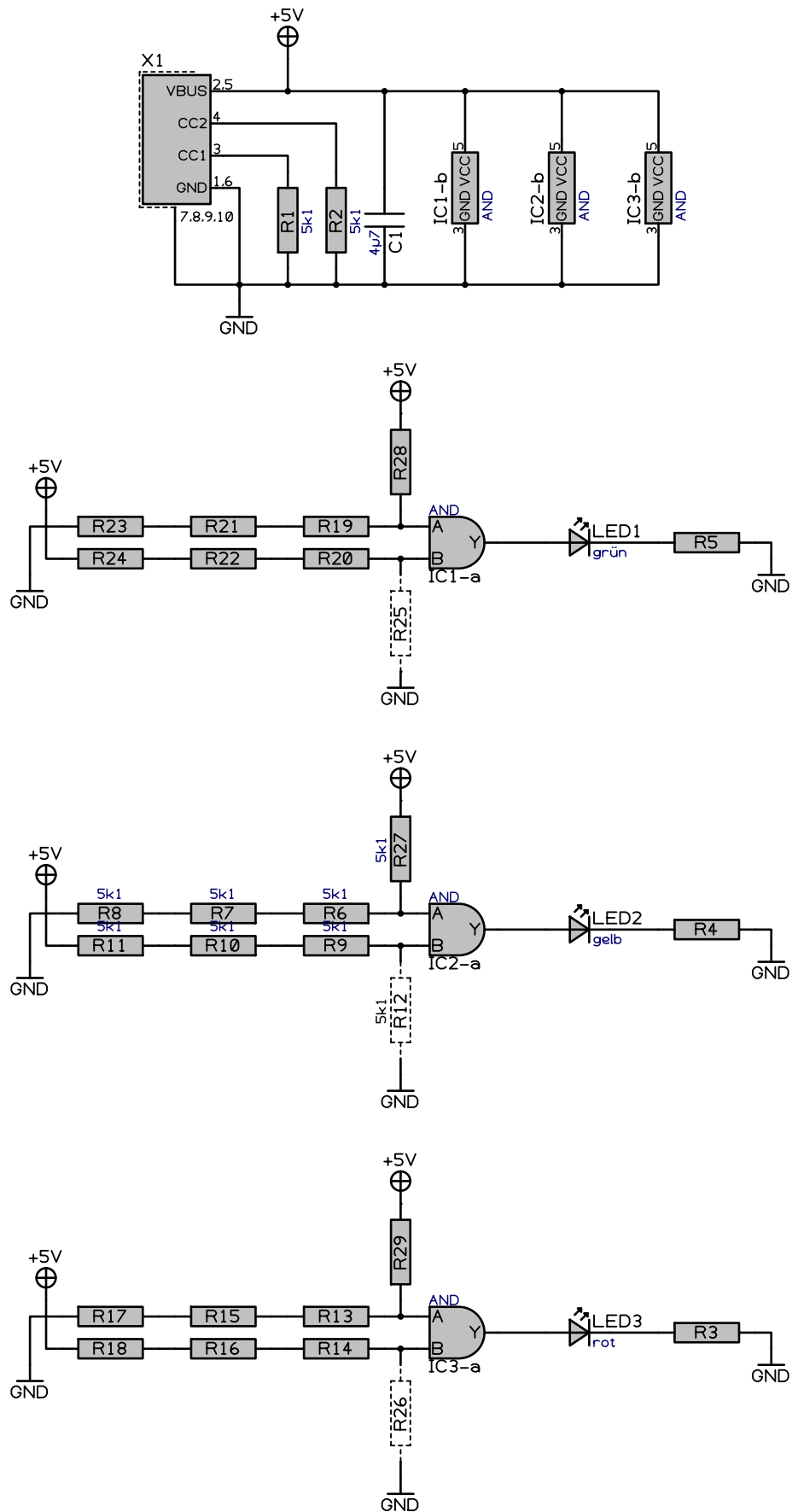


This soldering exercise is extremely challenging and demanding. Use a magnifying glass or microscope, a clean workspace, and high-quality tools!



3.2 Schematic

The schematic for the kit can be found on the next page, ensuring the correct scaling.



Autor hajo	Kunde meiner	Seitenname Page1	
eHaJo Elektronik Hannes Jochriem Oberfeldweg 12 83080 Oberaudorf Tel.: 08033 / 695 69 89 https://www.eHaJo.de	Dokumentenart Schaltplan	Dokumentenstatus frei gegeben	
	Titel, zusätzlicher Titel ise Incredible Solder Exercise 100	Sachnummer 0815	Gespeichert 17.01.2025
		Rev. 17	Blatt 1/1
		Spr. de	Größe A4